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SLAM Mapping of Information Fusion between Lidar and Depth Camera

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***Abstract*—In order to overcome the limitations of using a single sensor to locate and map the sterile mobile robot in complex environment, this paper combines the advantages of lidar and depth camera, presents a method to fuse the depth information of the depth camera with the lidar data to get point cloud data, uses gampping algorithm to construct two-dimensional indoor map, establishes the robot and simulation environment in gazebo, and tests the algorithm. Map information is more comprehensive than building maps directly using lidar.**

Keywords-lidar; kinect; SLAM; information fusion;

I. INTRODUCTION

Since the discovery of a new type of coronavirus in China at the end of 2019, sterilization is an arduous and extremely dangerous task [1]. Manpower alone will increase the risk of infection. Therefore, it is urgent to develop a sterilization service robot to replace the staff in medical sterilization and reduce the intensity and risk of related staff. Unlike traditional service robots, sterilizing robots may work in more complex and diverse environments, such as homes, hotels, restaurants, hospitals, and factories[2]. Due to the wide variety of environmental objects, there are too many uncertainties, and it is difficult to obtain comprehensive environmental characteristics with the information collected by a single sensor. For example, for single-line lidar, only point cloud data in the plane where the lidar is located can be scanned. For binocular vision or depth camera, although three-dimensional point cloud data can be obtained, its observation distance is close and is susceptible to environmental factors such as lighting. Therefore, the perception of such a complex environmental robot needs to be achieved through a variety of sensors, and now the application of fusion perception technology needs to be strengthened [3-4]. Combining the advantages of lidar and depth camera, this paper fuses two-dimensional lidar and depth camera point cloud data to make the map closer to the actual environment. Setting up the robot and simulation environment in gazebo and testing the algorithm can get more comprehensive map information than building maps directly using lidar.

II. SENSOR DATA COLLECTION AND PROCESSING

A. Lidar data

Triangular ranging is commonly used in lidar acquisition. The principle is that the laser signal reflects when it illuminates the surface of the measured object at a certain angle of incidence. The incident and reflected light form a triangle, which is then received by the lidar internal acquisition system. Based on the triangular relationship, the distance and relative azimuth of the target object to the radar can be calculated in real time by the lidar internal processor[5]. The implementation is shown in Fig. 1.

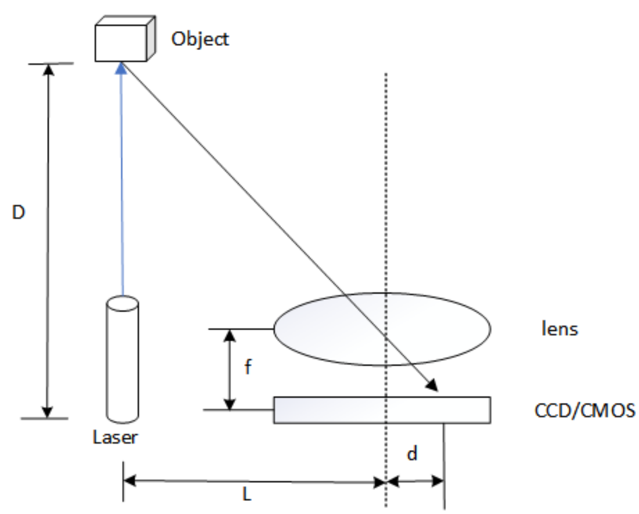

Figure 1. Lidar triangulation schematic

Laser reflects a laser from an Object with a distance of D in the direction of the laser. Receiving laser is usually a CCD or CMOS. The laser reflected by the object is captured by a CCD or CMOS through lens "pinhole imaging". The focal length is f, the distance from the main optical axis of the lens to the point projected onto the CCD or CMOS is d, and the distance between

Laser and the main optical axis is L. You can see from Fig. 1 that：

$$D = \frac{f*(L+d)}{d} \quad (1)$$

Thus, the lidar and target distances can be calculated. The lidar distance data published in the ROS system is represented by polar coordinates, and an ordered two-dimensional array can be obtained for environmental scanning. The lidar scan range is [0°,360°], and 360 polar points are scanned.

$$T_i = (\rho_i, \theta_i) \quad (2)$$

With formula (3), polar coordinate point data can be converted to rectangular coordinate system.

$$\begin{cases} x_i = \rho_i \cos\theta_i \\ y_i = \rho_i \sin\theta_i \end{cases} \quad (3)$$

B. Depth camera data

Depth camera data processing mainly converts the depth map data to simulated lidar data, and makes the three-dimensional data two-dimensional, so as to fuse with the lidar data to build the map[6]. Depthimage_to_laserscan function pack in ROS system is used for depth image 2D, and its principle is shown in Fig. 2.

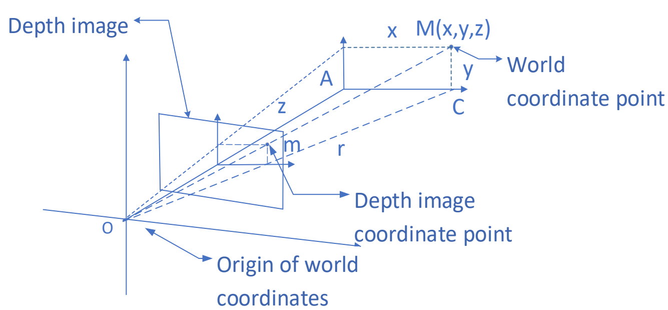

Figure 2. Depth map to laser principl

(1) Calculates the position coordinate M (x, y, z) of the point m in the image coordinate system in the world coordinate system, as shown in formula (4).

$$\begin{cases} x = z_c \times (u - u_0) \times dx/f \\ y = z_c \times (v - v_0) \times dy/f \\ z = z_c \end{cases} \quad (4)$$

Formula (4), (u, v) is a coordinate point in the image coordinate system, (u_0, v_0) is the center point of the image coordinate system, (x, y, z) represents the coordinate of the point in the world coordinate system, where z_c represents the distance from the target to the camera and f is an in-camera parameter.

(2) Calculates the angle AOC between the straight line AO and the straight line CO, as shown in formula (5).

$$\theta = \tan^{-1}\frac{x}{z} \quad (5)$$

(3) Projects the angular AOC to the corresponding laser scanning range, and the laser scanning range is $[\alpha, \beta]$, if the laser beam is divided into N parts, the array Laser [N] can be used to represent $[\alpha, \beta]$ Laser data in range. Thus, the index value n projected from point M to the array Laser can be shown in formula (6).

$$n = \frac{\theta - \alpha}{\frac{\beta - \alpha}{N}} = \frac{N(\theta - \alpha)}{\beta - \alpha} \quad (6)$$

Laser[n] has a value of r from point C projected by M on the x-axis to the origin of the world coordinates, as shown in formula (7).

$$\text{Laser}[n] = r = OC = \sqrt{x^2 + z^2} \quad (7)$$

Therefore, when converting depth data to lidar data using the lidar data format as standard, by taking the minimum depth value of the depth image at an integer angle in the horizontal range as the lidar distance data, it is equivalent to taking the horizontal projection of the three-dimensional depth data and adding the corresponding angle value, thus completing the process of converting depth data to lidar data, and then saving the data to an array obtained by the lidar sensor. The acquisition of distance information from the depth camera to the front obstacle has been completed.

This completes the conversion of camera depth data to lidar data, save the minimum values of each column on the depth image in turn to the radar Laser [N] array. Theoretically, we will get the closest obstacle distance vertically in front of the camera, and the Laser[N] array represents the distance from each obstacle point horizontally to the camera. In practice, however, the height of the vertical sweep surface is very demanding. Because when the scan_height scan range exceeds a certain value, the ground is introduced into the array as an obstacle, and the consequence is that the obstacle will be scanned globally, so a suitable scan_height needs to be found and the scan_height will be as small as possible. Therefore, the vertical information will be lost a lot, so in order to get more comprehensive data, it is necessary to make the best use of the plane data where the depth camera is located, so the installation position of the depth camera in the robot needs to be as far down as possible so that the robot can detect and map lower-height obstacles by turning the depth camera into lidar data.

III. Lidar and Depth Camera Data Fusion

Since lidar can only obtain point cloud data on the mounting plane, when lidar is used alone, information about obstacles under the mounting plane will be unavailable, and a lot of information will be lost when building a map. When using point cloud data from depth camera to lidar alone to build a map, because the depth camera itself has a smaller field of view and therefore a smaller point cloud data range, more time is needed to collect point cloud data when building the map. Because there is less data in itself, the map will also be elegant and so on, which makes it impossible to get a better map, so Lidar-Based map is required. Build a map after fusing point cloud data from a deep camera. The specific flow is shown in Fig. 3.

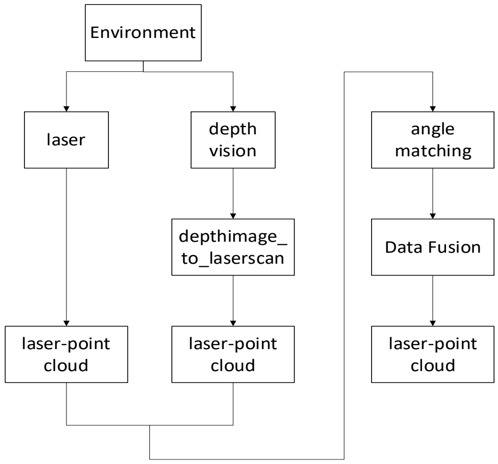

Figure 3. Laser and visual point cloud data fusion

Integrate depth camera and lidar into the robot, where the higher position of the lidar robot is used to collect global point cloud data, and the lower position of the robot is used to detect point cloud data with lower height obstacles. After collecting depth image, the depth camera is converted to LIDAR point cloud data through depthimage_to_laserscan function pack. Angle matching is required after converting to radar data. Generally, the acquisition angle of lidar is [0°,360°]. When one degree of data is collected, 360 polar coordinate points will be scanned. In the array range [], each data is the distance between point cloud and pole collected by corresponding radian value, while the acquisition of point cloud data by depth camera is related to the horizontal field of view of camera internal parameters. For example, the horizontal field of view is θ As shown in Fig. 4, ∠ COD is the angle of view of the camera θ, Its corresponding radian is shown in Formula (8), where Rad is the horizontal field of view θ The radian value corresponding to the camera of.

$$\text{Rad} = \frac{\theta \times Pi}{180} \tag{8}$$

If the depth camera is installed in the middle of the robot, that is, its horizontal field of view is in the positive direction of the x-axis, that is, the radian value of the axis where the dotted line is located in Fig. 4 is 0, so the radian of the axis where OC is located is Rad/2 relative to the principal axis, and the radian of the axis where OD is located is Rad/2 relative to the principal axis, then when the depth camera is installed in the middle of the robot, its radian range is shown in Formula (9).

$$[-Rad/2, Rad/2] \tag{9}$$

After the effective angle of action is obtained from the radar and camera attributes, the angle matching is completed and data fusion can be carried out. Traverse the corresponding radian range of the camera's horizontal field of view to determine the distance between the LIDAR point cloud data in corresponding radians and the depth camera point cloud data, such as the lidar distance is greater than the depth camera distance, that is, change the lidar data in this radian to the depth camera data. It is equivalent to completing the information lost when lidar collects data.

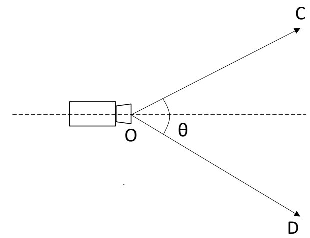

Figure 4. Camera horizontal field angle

Finally, the topic of lidar data after fusion is republished for gmapping Feature Pack subscription and mapping.

IV. Experimental Simulation

A. Robot Modeling

In this paper, a mobile robot model is built by itself in the gazebo simulation environment. To save modeling time, the robot is built using solidworks[7]. After importing into ROS, modify the URDF file to add lidar and lidar brackets and depth cameras. The final robot display in the gazebo simulation environment is shown in Fig.5. The red part is the radar stand, the yellow part is the lidar, and the blue part under the robot is the depth camera.

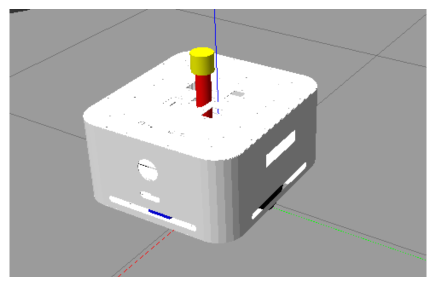

Figure 5. Robot model in gazebo

B. Setup of simulation environment

A simple simulation environment has been built in gazebo. As shown in Fig. 6, there are three objects in the environment that cannot be detected by lidar. The height of the objects labeled with red circles in the diagram is overall low, that is, the lidar cannot scan. Others, such as cylinders and cubes, can be detected by lidar.

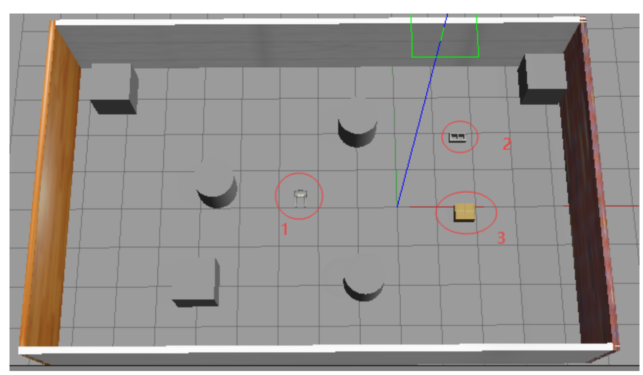

Figure 6. Simulation environment

C. gmapping Building Maps

1) Lidar Mapping

Using lidar alone to build maps, lidar parameters are sampled at [0°,360°], the sampling points are 360, the minimum sampling distance is 0.1m, the maximum sampling distance is 30m, and the update frequency is 5.5hz. Fig.7 is a lidar scan in a simulation environment. The blue color in the figure is a lidar visual scan beam. You can see that the lidar did not scan the three red circle labeled objects in Fig.6

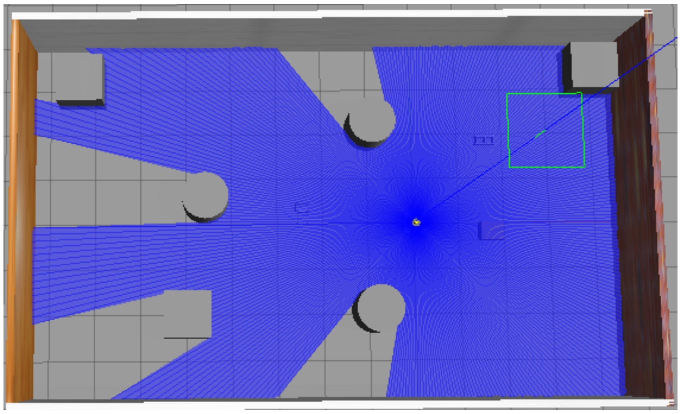

Figure 7. Lidar scanning in simulation environment

After running the gmapping node, use teleop_twist_ keyboard node operates the robot to move and build the map, and the final saved map is shown in Fig.8

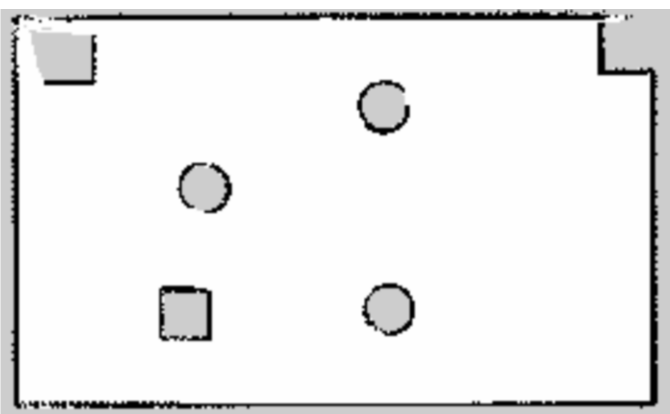

Figure 8. Lidar building maps

The resulting map shows that lidar alone cannot detect and build three low objects circled in red in Fig. 6.

2) Post-construction of lidar and depth visual data fusion

Using lidar and depth visual fusion to build a map, first converts the depth information into pseudo lidar information using depthimage_to_laserscan function pack, fuses the real lidar information with pseudo lidar information through the fusion process, generates a new lidar data, and publishes a scan topic to subscribe to the gmapping function pack. The final map built is shown in Fig. 9. You can see that although the detected boundary has some burrs, the 1, 2, 3 objects labeled with red circles in Fig. 6 have been detected.

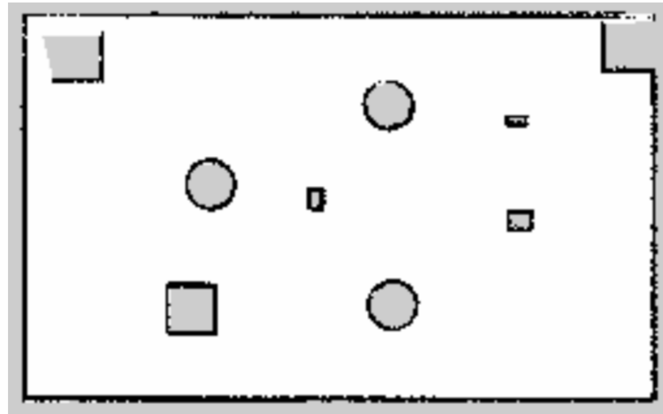

Figure 9. Construction site after lidar and visual fusion

The resulting map after data fusion preserves the high accuracy of lidar mapping, while detecting obstacles that are lower than the height of the radar installation and obtaining a more complete map.

V. CONCLUSIONS

This paper presents a data fusion method which combines the advantages of lidar and depth camera to solve the problem that the working environment of sterilizing robot is complex and incomplete. Then a map is constructed using gmapping algorithm. The feasibility and validity of this method are verified through theoretical derivation and simulation. The map information constructed by this method is more comprehensive than that by single sensor.

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